Singularity in Bohemian Upper-Hessenberg Matrices: Anatomy and Geometric Rate

Kevin Josué Martínez Herrera
Escuela de Física, Universidad de Costa Rica
Email: kevin.martinezherrera@ucr.ac.cr

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Introduction

$$H = \begin{pmatrix} h_{11} & h_{12} & h_{13} & \cdots & h_{1n} \\ s_1 & h_{22} & h_{23} & \cdots & h_{2n} \\ 0 & s_2 & h_{33} & \cdots & h_{3n} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & s_{n-1} & h_{nn} \end{pmatrix}$$

$$s_i = \begin{cases} 1, & \text{normalized model} \\ \sim \text{Unif}\{0,\ldots,m-1\}, & \text{free model} \end{cases} \qquad i = 1,\ldots,n-1$$

- A matrix is singular when the determinant is zero. Then, it collapses information, some direction of space is squashed to zero, and the matrix can't be inverted. Whether a random matrix is singular, and why, is a classic question in probability and numerical linear algebra [1].
- We study upper Hessenberg matrices: almost upper-triangular, but with one extra nonzero diagonal just below the main one. This shape arises naturally in the Arnoldi algorithm, a core tool for reducing large linear-algebra problems to a manageable size [2].
- We make the matrix Bohemian (BOunded HEight Matrix of Integers) [3,4] drawing every free entry from a small fixed set {0,…,m−1} instead of the real numbers. This turns singularity into a discrete, combinatorial question.
- Two natural models emerge, depending on how the subdiagonal is treated: Free and Normalized
- The purpose is to establish how likely is such a matrix to be singular, and if so, to also determine the the likeliness for said singularity to be structural, or a genuine arithmetic event

Methodology

- **Exact integer arithmetic:** Determinant is computed via the continuant recurrence $d_k = \sum_{j=1}^{k} (-1)^{k-j} h_{j,k} d_{j-1}$, evaluated directly over $\mathbb{Z}$ avoiding floating-point rounding [5]; singularity $d_n = 0$ is detected exactly, also reducing determinant evaluation from $O(n^3)$ to $O(n^2)$, enabling large-scale sampling. Results cross-checked against Bareiss integer Gaussian elimination [6].
- **Exhaustive enumeration:** For small (n, m), $mn(n+1)/2$ matrices were enumerated directly, yielding exact rational singularity probabilities as a ground-truth check on sampled estimates
- **Rao–Blackwell estimation of** $\gamma(m)$: Writing $d_n = \mp h_{1,n} + R$ with R independent of $h_{1,n}$, we integrate $h_{1,n}$ analytically:

$$P(d_n = 0 | \text{rest}) = \frac{1}{m} \vec{1}\{\cdot\} \in \{0, 1/m\}$$

rather than $\{0, 1\}$. This cuts estimator variance by a factor m, allowing tight bootstrap confidence intervals on the geometric base $\gamma(m)$.

- **Lyapunov exponent estimation:** $\lambda(m) = \lim_{k\to\infty} \frac{1}{k} \log |d_k|$

estimated via per-step rescaling up to $n = 1000$, isolating the growth-rate component of $\gamma(m) = e^{-\lambda} \rho(m)$ from the arithmetic-resonance factor $\rho(m)$.

- **Condition-number analysis:** Singular values computed via SVD on batches of sampled matrices, with the exact-singularity flag (from integer recurrence) used to separate the $k = \infty$ atom from the continuous distribution, avoiding floating-point contamination of the bulk.

Bibliography

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Spectral Structure Analysis

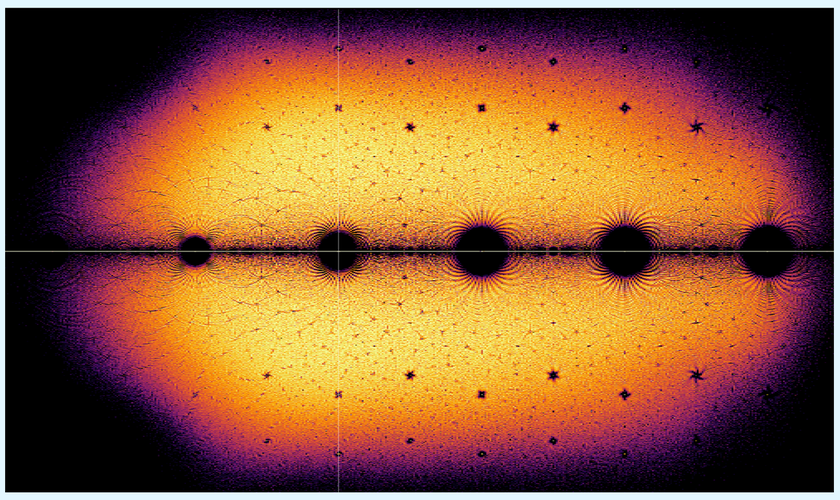


Figure 1. *Empirical Eigenvalue Density over the complex plane for the Bohemian upper-Hessenberg family, n = 5, alphabet {0,1,2,3} (m = 4), obtained with 100 million Monte Carlo samples, accumulated via streaming histograms*

Bohemian matrices show fractal-like eigenvalue patterns with integer-driven voids. More voids and wider support as m grows, denser bulk as n grows. This project isolates the origin and tracks how its probability evolves with n and m.

Free and Normalized Model Analysis

For the free model, the singularity is structural. A single vanishing subdiagonal entry $s_i = 0$ with $P(\text{singular}) \approx 1/m$ splits H into blocks and forces it to be singular. The dominant cause is thus the subdiagonal: $P(\text{singular}) \approx P(\exists s_i = 0) \approx (n-1)/m$, which grows with dimension (Fig. 3). 95–99% of singular free matrices carry such a zero, and the corank can exceed 1. The genuine phenomenon survives only with the subdiagonal pinned to unity, hence the normalized model becomes the object of interest, for which the following is derived:

Theorem 1. (Corank at most one)

Every normalized Bohemian upper-Hessenberg matrix satisfies $\text{rank}(H) \geq n - 1$; hence a singular H has a kernel of dimension exactly 1.

Theorem 2. (Singularity Probability Bound)

$P(\text{singular}) \leq 1/m$ and Geometric decay $P = \Theta(\gamma(m)^n)$

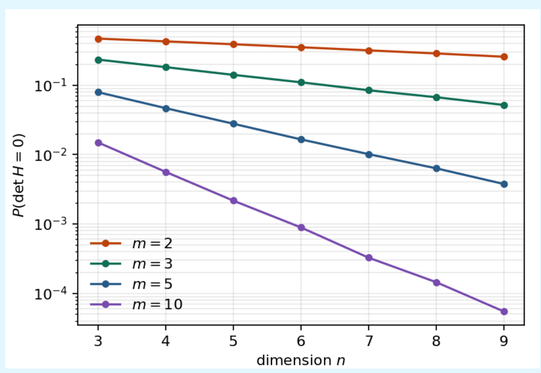


Figure 2. *Decay of the singularity probability for the normalized model (semilog scale)*

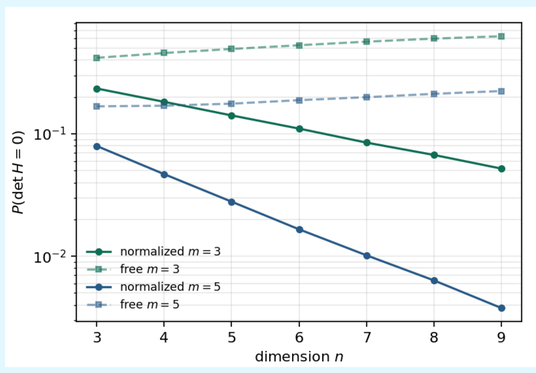


Figure 3. *Decay of the normalized model vs growth of the free model (semilog scale)*

Corollary 1 (Genuine Dominance)

Conditioned on being singular, the matrix is almost surely singular for a genuine arithmetic reason, not a structural one.

Conjecture 1 (Existence of the Geometric Rate)

For $m \geq 2$, the limit $\gamma(m) = \lim_{n\to\infty} P(d_n = 0)^{1/n}$ exists and satisfies the decomposition $\gamma(m) = e^{-\lambda(m)} \rho(m)$, where $\lambda(m)$ is the Lyapunov exponent for the minor sequence and $\rho(m) \geq 1$ is an arithmetic-resonance factor.

Conditioning Number κ Analysis

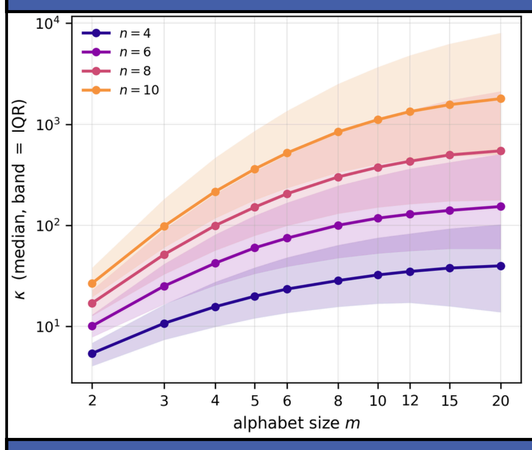


Figure 4. *Median κ versus alphabet size (band: interquartile range)*

Distribution shifts toward larger κ as m grows, so (n, m) controls the conditioning behind classical Krylov solvers (Arnoldi [2], GMRES [7])

It also relates to the study of open quantum systems. Krylov analysis reduces the Lindbladian to upper-Hessenberg form. That said, this is not a specific physical model, but a shared structural setting, one of the same kind behind non-Hermitian Krylov reductions [8].

Conclusions

The unit subdiagonal caps the corank at 1, forcing structural singularity to $O(m^{-n})$, so genuine dependence dominates unconditionally, unlike the dense i.i.d. case.

The continuant recurrence gives $P(\text{singular}) \leq 1/m$, attained at n = m = 2, with geometric decay $\Theta(\gamma(m)^n)$ far above the dense 1/m. Existence of $\gamma(m)$ remains conjectural, since each step depends on prior minors.

The free model relapses into structural singularity, confirming the dense upper triangle, not the shape alone, drives the phenomenon.

κ grows exponentially in n at an m-dependent rate, serving as a null-model baseline for Krylov and open-quantum reductions, opening a path for future projects.